



MUHAMMAD ISHAQ

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SUMMARY

AI Engineer focused on building smart systems powered by deep learning, language understanding, and Generative AI. Skilled in creating reliable RAG pipelines, scalable AI workflows, and production-ready models. Strong background in conversational tools with clear answers, multilingual features, and smooth voice output, delivering complete AI solutions from idea to launch.

WORK EXPERIENCE

AI Engineer, Ventive LLC Boise (Remote)

Nov 2025- March 2026

- Built and deployed scalable AI and data systems at Ventive, including NLP-powered intelligent chatbots and automated data pipelines on AWS.
- Designed and implemented cloud-based data architectures using AWS services such as AWS Glue and Bedrock with Docker-based deployments for production environments.
- Developed and optimized ETL workflows and data pipelines to enable efficient processing of large-scale enterprise data for analytics and AI use cases.
- Engineered knowledge graphs to model complex entity relationships, reducing AI retrieval latency by up to 85% while maintaining high accuracy.

Team Lead AI & Data Science, Octans Digital Islamabad

Sep 2024- May 2025

- Led AI/Data Science team at Octans Digital (Islamabad), driving development of scalable Generative AI and SaaS-based applications.
- Built and deployed RAG pipelines using LangChain and AWS Bedrock, achieving 99.9% uptime and sub-second retrieval latency.
- Designed inference optimisation strategies that reduced operational costs by 25% while improving model throughput and efficiency.
- Mentored junior engineers and apprentices, establishing structured development practices for GenAI-based SaaS products.
- Strong communicator with experience collaborating with cross-functional teams including engineers, product managers, and stakeholders to deliver AI-driven solutions.

Associate AI Engineer, Octans Digital Islamabad

Mar 2023 - Sep 2024

- Worked as an AI Engineer, designing and deploying scalable Generative AI solutions using state of the art LLM models and AWS SageMaker for model fine-tuning and inference.
- Built multilingual NLP systems and computer vision pipelines for real-world applications, focusing on automation, accuracy, and production scalability.
- Developed AI-powered chatbots, speech-to-text pipelines, and intelligent document processing systems to streamline enterprise workflows.
- Delivered production-grade AI solutions that improved operational efficiency and enabled intelligent decision-making across business processes.
- Implemented end-to-end ML workflows including data processing, model training, deployment, and monitoring using cloud-native and containerized environments (ECS, EKS) .
- Actively participated in technical discussions, architecture design reviews, and sprint planning to improve system design and delivery efficiency.

EDUCATION

Bachelors of Computer Science, HITEC UNIVERSITY TAXILA

Sep 2017- July 2021

- **Final year project:** Developed a defense-sector automation system to replace manual inventory management with a web-based software solution using C# and ASP.NET Web Forms.
- Designed and implemented an automated Inventory Management System to digitize data entry, tracking, and retrieval processes, improving operational efficiency and reducing manual workload.
- Built a user-friendly interface for secure and structured inventory handling tailored to defense requirements.
- Developed an object detection-based automated attendance system to reduce manual attendance marking and improve accuracy.
- Integrated computer vision techniques to enable real-time identification and automated record generation for attendance automation use case.
- Delivered an end-to-end system covering requirement analysis, system design, software development, and automation of legacy manual workflows.

Masters of Computer Science, FAST UNIVERSITY ISLAMABAD

Apr 2022- Aug 2024

- **MS Thesis:** Conducted thesis on Fabric DefectAI, Deep Learning and Computer Vision for Automated Fabric Fault Detection using state-of-the-art object detection and transformer-based models.
- Developed a custom-annotated dataset for fabric defect detection and structured it for training deep learning models.
- Trained and fine-tuned YOLOv8 object detection model with hyperparameter optimization to improve detection accuracy and robustness.
- Applied advanced data augmentation techniques to enhance dataset diversity and improve model generalization.
- Integrated Vision Transformers (ViTs) with CNN-based pipelines to improve feature representation and overall model performance.
- Delivered an end-to-end computer vision system for automated defect detection, improving accuracy and reducing manual inspection effort.

CERTIFICATES

- AWS Certified Cloud Practitioner
- Data Science Methodology – IBM Tools for Data Science
- IBM Introduction to Containers with Docker and Kubernetes
- Generative AI with Large Language Models
- AWS Bedrock ChatGPT Prompt Engineering for Developers

ADDITIONAL INFORMATION

- Programming Languages: Python, Nodejs
- Data Visualization tools: PowerBI, Tableau
- Languages: English, Urdu, Pushto